

Australian statement of hazardous nature : Classified as hazardous according to criteria of Safe Work Australia

## Section 1 - Identification

**Product Name** Benzyl n-butyl phthalate

**CAS No** 85-68-7

**Product Code** B24769

**Address** ThermoFisher Scientific Australia Pty Ltd  
 5 Caribbean Drive, Scoresby  
 VICTORIA 3179, Australia

**Emergency Tel.** CHEMTREC®  
 03 9757 4559 or +613 9757 4559

**Telephone / Fax Numbers** Tel: 1300 735 292  
 Fax: 1800 067 639

**E-mail address** ANZinfo@thermofisher.com

**Recommended Use** Laboratory chemicals.

**Uses advised against** This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.  
 This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction. This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern.

## Section 2 - Hazard(s) Identification

### Classification under Safe Work Australia

Classified as hazardous according to criteria of Safe Work Australia

**Physical hazards**  
 No hazards identified

### **Health hazards**

Reproductive Toxicity Category 1B

### **Environmental hazards**

Acute aquatic toxicity Category 1  
 Chronic aquatic toxicity Category 1

### Label Elements



Health Hazard



Environment

**Signal Word****Danger****Hazard Statements**

H360 - May damage fertility or the unborn child

H410 - Very toxic to aquatic life with long lasting effects

**Precautionary Statements**

P201 - Obtain special instructions before use

P202 - Do not handle until all safety precautions have been read and understood

P264 - Wash face, hands and any exposed skin thoroughly after handling

P280 - Wear eye protection/ face protection

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P308 + P313 - IF exposed or concerned: Get medical advice/attention

P403 - Store in a well-ventilated place

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other information**

Contains a known or suspected endocrine disruptor

Included in the list established in accordance with Article 59(1) for having endocrine disrupting properties

Contains a substance on the National Authorities Endocrine Disruptor Lists

**Section 3 - Composition and Information on Ingredients**

| Component              | CAS No  | Weight % |
|------------------------|---------|----------|
| Butyl benzyl phthalate | 85-68-7 | >95      |

**Section 4 - First Aid Measures**

|                                            |                                                                                                                                                  |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>                          | Remove to fresh air. If breathing is difficult, give oxygen. Get medical attention.                                                              |
| <b>Ingestion</b>                           | Do NOT induce vomiting. Get medical attention.                                                                                                   |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. Get medical attention.                                                        |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                  |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                             |
| <b>Most important symptoms and effects</b> | No information available.                                                                                                                        |
| <b>Notes to Physician</b>                  | Treat symptomatically.                                                                                                                           |

**Section 5 - Fire Fighting Measures****Suitable Extinguishing Media**Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Hazardous Decomposition Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).

**Specific Hazards Arising from the Chemical**

Do not allow run-off from fire-fighting to enter drains or water courses.

**Special protective equipment and precautions for fire fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## Section 6 - Accidental Release Measures

**Emergency procedures**

Ensure adequate ventilation. Use personal protective equipment as required.

**Environmental Precautions**

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

**Methods for Containment and Clean Up****Clean-up methods - small spillage**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

**Clean-up methods - large spillage**

Typically only supplied in small quantities as packaged goods.

If extremely toxic or used in larger quantities ensure a spillage action plan is in place. Evacuate area. Control the source and/or contain the spill if safe and able to do so. Use temporary diking, sand bags, dry sand, earth or proprietary booms/absorbent pads if available. Obtain advice on containment, neutralisation and clean-up from local emergency responders.

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

**Precautions for Safe Handling**

Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Do not breathe (dust, vapor, mist, gas). Do not ingest. If swallowed then seek immediate medical assistance.

**Conditions for Safe Storage, Including any Incompatibilities**

Keep containers tightly closed in a dry, cool and well-ventilated place.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

## Section 8 - Exposure Controls and Personal Protection

**Exposure limits**

**UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020. **DE** - MAK and BAT values of Hazardous Chemical Compounds in the Work Area. Published by German Research Foundation on July 1, 2011

| Component              | Australia | New Zealand WEL          | ACGIH TLV | The United Kingdom                                                 | Germany                                                                                        |
|------------------------|-----------|--------------------------|-----------|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Butyl benzyl phthalate |           | TWA: 5 mg/m <sup>3</sup> |           | STEL: 15 mg/m <sup>3</sup> 15 min<br>TWA: 5 mg/m <sup>3</sup> 8 hr | TWA: 20 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>TWA: 20 mg/m <sup>3</sup> (8 |

|  |  |  |  |  |                                                  |
|--|--|--|--|--|--------------------------------------------------|
|  |  |  |  |  | Stunden). MAK<br>Höhepunkt: 40 mg/m <sup>3</sup> |
|--|--|--|--|--|--------------------------------------------------|

**Biological limit values**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

**Exposure Controls****Engineering Measures**

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Wear safety glasses with side shields (or goggles) (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time | Glove thickness | AUS/NZ Standard | Glove comments        |
|----------------|-------------------|-----------------|-----------------|-----------------------|
| Nitrile rubber | See manufacturers | -               | AS/NZS 2161     | (minimum requirement) |
| Neoprene       | recommendations   |                 |                 |                       |
| Natural rubber |                   |                 |                 |                       |
| PVC            |                   |                 |                 |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure

**Respiratory Protection**

Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

**Recommended Filter type:**

Organic gases and vapours filter Type A Brown conforming to EN14387 (or AUS/NZ equivalent)

**Recommended half mask:-**

Valve filtering: EN405 or Half mask: EN140 plus filter, EN 141 (or AUS/NZ equivalent)  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls**

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

## Section 9 - Physical and Chemical Properties

**Information on basic physical and chemical properties****Appearance**

Colorless to yellow

**Physical State**

Viscous liquid Liquid

**Odor**

Practically odorless

**Odor Threshold**

No data available

**pH**

No information available

|                                                |                          |                                          |
|------------------------------------------------|--------------------------|------------------------------------------|
| <b>Melting Point/Range</b>                     | -35 °C / -31 °F          |                                          |
| <b>Softening Point</b>                         | No data available        |                                          |
| <b>Boiling Point/Range</b>                     | 370 °C / 698 °F          | 760 mmHg                                 |
| <b>Flash Point</b>                             | 215 °C / 419 °F          | <b>Method</b> - No information available |
| <b>Evaporation Rate</b>                        | No data available        |                                          |
| <b>Flammability (solid,gas)</b>                | Not applicable           | Liquid                                   |
| <b>Explosion Limits</b>                        | No data available        |                                          |
| <b>Vapor Pressure</b>                          | No data available        |                                          |
| <b>Vapor Density</b>                           | No data available        | (Air = 1.0)                              |
| <b>Specific Gravity / Density</b>              | 1.1                      |                                          |
| <b>Bulk Density</b>                            | Not applicable           | Liquid                                   |
| <b>Water Solubility</b>                        | 2.7 mg/L (25°C)          |                                          |
| <b>Solubility in other solvents</b>            | No information available |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                                          |
| <b>Component</b>                               | <b>log Pow</b>           |                                          |
| Butyl benzyl phthalate                         | 4.91                     |                                          |
| <b>Autoignition Temperature</b>                | No data available        |                                          |
| <b>Decomposition Temperature</b>               | No data available        |                                          |
| <b>Viscosity</b>                               | No data available        |                                          |
| <b>Explosive Properties</b>                    | No information available |                                          |
| <b>Oxidizing Properties</b>                    | No information available |                                          |

**Other information**

|                          |            |
|--------------------------|------------|
| <b>Molecular Formula</b> | C19 H20 O4 |
| <b>Molecular Weight</b>  | 312.37     |

## Section 10 - Stability and Reactivity

|                                         |                                                          |
|-----------------------------------------|----------------------------------------------------------|
| <b>Reactivity</b>                       | None known, based on information available               |
| <b>Stability</b>                        | Stable under normal conditions.                          |
| <b>Conditions to Avoid</b>              | Incompatible products, Excess heat.                      |
| <b>Incompatible Materials</b>           | Strong oxidizing agents.                                 |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ). |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                 |

## Section 11 - Toxicological Information

**Information on Toxicological Effects****Product Information****(a) acute toxicity;**

|                   |                                                                  |
|-------------------|------------------------------------------------------------------|
| <b>Oral</b>       | Based on available data, the classification criteria are not met |
| <b>Dermal</b>     | Based on available data, the classification criteria are not met |
| <b>Inhalation</b> | Based on available data, the classification criteria are not met |

| Component              | LD50 Oral                 | LD50 Dermal               | LC50 Inhalation             |
|------------------------|---------------------------|---------------------------|-----------------------------|
| Butyl benzyl phthalate | LD50 = 2330 mg/kg ( Rat ) | LD50 = 6700 mg/kg ( Rat ) | LC50 > 6.7 mg/L ( Rat ) 4 h |

**(b) skin corrosion/irritation;** No data available

**(c) serious eye damage/irritation;** No data available

**(d) respiratory or skin sensitization;**

Respiratory No data available  
Skin No data available

**(e) germ cell mutagenicity;** No data available

**(f) carcinogenicity;** No data available

There are no known carcinogenic chemicals in this product

**(g) reproductive toxicity;**  
**Reproductive Effects**  
**Developmental Effects**

Category 1B  
Possible risk of impaired fertility  
May cause harm to the unborn child

**(h) STOT-single exposure;** No data available

**(i) STOT-repeated exposure;** No data available

**Target Organs** No information available.

**(j) aspiration hazard;** No data available

**Other Adverse Effects** See actual entry in RTECS for complete information

**Symptoms / effects, both acute and delayed** No information available

## Section 12 - Ecological Information

**Ecotoxicity effects**

Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component              | Freshwater Fish                                                                    | Water Flea                                                                                                                                                                                                          | Freshwater Algae                                                                                                                    | Microtox |
|------------------------|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|----------|
| Butyl benzyl phthalate | Lepomis macrochirus:<br>LC50=1.7 mg/L 96h<br>Salmo gairdneri:<br>LC50=1.1 mg/L 96h | EC50: = 0.97 mg/L, 48h<br>(Daphnia magna)<br>EC50: = 1.28 mg/L, 48h<br>semi-static (Daphnia magna)<br>EC50: 0.9 - 1.1 mg/L, 48h<br>Static (Daphnia magna)<br>EC50: > 0.76 mg/L, 48h<br>Flow through (Daphnia magna) | EC50: 0.2 - 28.2 mg/L, 72h<br>(Pseudokirchneriella subcapitata)<br>EC50: 0.02 - 0.25 mg/L, 96h<br>(Pseudokirchneriella subcapitata) |          |

**Persistence and Degradability**

Readily biodegradable

**Persistence**

May persist, based on information available.

**Degradation in sewage treatment plant**

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Bioaccumulative Potential**

May have some potential to bioaccumulate

| Component              | log Pow | Bioconcentration factor (BCF)                               |
|------------------------|---------|-------------------------------------------------------------|
| Butyl benzyl phthalate | 4.91    | 255 dimensionless<br>194 dimensionless<br>9.4 dimensionless |

**Mobility**

The product is water soluble, and may spread in water systems. : Will likely be mobile in the environment due to its water solubility Highly mobile in soils

**Endocrine Disruptor Information**

| Component | EU - Endocrine Disruptors | EU - Endocrine Disruptors - | Japan - Endocrine Disruptor |
|-----------|---------------------------|-----------------------------|-----------------------------|
|-----------|---------------------------|-----------------------------|-----------------------------|

|                                     | Candidate List                                                 | Evaluated Substances  | Information |
|-------------------------------------|----------------------------------------------------------------|-----------------------|-------------|
| Butyl benzyl phthalate              | Group I Chemical                                               | High Exposure Concern |             |
| <b>Persistent Organic Pollutant</b> | This product does not contain any known or suspected substance |                       |             |
| <b>Ozone Depletion Potential</b>    | This product does not contain any known or suspected substance |                       |             |

## Section 13 - Disposal Considerations

|                                            |                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations. |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point.                                                                                                                                                                                                                                            |
| <b>Other Information</b>                   | Chemical wastes should be disposed through a licensed commercial waste collection service. Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not let this chemical enter the environment.                      |

## Section 14 - Transport Information

### IMDG/IMO

|                                |                                                      |
|--------------------------------|------------------------------------------------------|
| <b>UN-No</b>                   | UN3082                                               |
| <b>Proper Shipping Name</b>    | Environmentally hazardous substances, liquid, n.o.s. |
| <b>Technical Shipping Name</b> | Butyl benzyl phthalate                               |
| <b>Hazard Class</b>            | 9                                                    |
| <b>Packing Group</b>           | III                                                  |

| Component                                | IMDG Marine Pollutant                                 |
|------------------------------------------|-------------------------------------------------------|
| Butyl benzyl phthalate<br>85-68-7 (>95 ) | IMDG regulated marine pollutant (Listed in the index) |

### ADG

|                                |                                                      |
|--------------------------------|------------------------------------------------------|
| <b>UN-No</b>                   | UN3082                                               |
| <b>Proper Shipping Name</b>    | Environmentally hazardous substances, liquid, n.o.s. |
| <b>Technical Shipping Name</b> | Butyl benzyl phthalate                               |
| <b>Hazard Class</b>            | 9                                                    |
| <b>Packing Group</b>           | III                                                  |

### IATA

|                                |                                                      |
|--------------------------------|------------------------------------------------------|
| <b>UN-No</b>                   | UN3082                                               |
| <b>Proper Shipping Name</b>    | Environmentally hazardous substances, liquid, n.o.s. |
| <b>Technical Shipping Name</b> | Butyl benzyl phthalate                               |
| <b>Hazard Class</b>            | 9                                                    |
| <b>Packing Group</b>           | III                                                  |

|                               |                                                                                                          |
|-------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>Environmental hazards</b>  | Dangerous for the environment<br>Product is a marine pollutant according to the criteria set by IMDG/IMO |
| <b>Special Precautions</b>    | No special precautions required                                                                          |
| <b>Additional information</b> | None known                                                                                               |

## Section 15 - Regulatory Information

**Safety, health and environmental regulations/legislation specific for the substance or mixture****National Regulations****Australia**

See section 8 for national exposure control parameters.

**Standard for the Uniform Scheduling of Medicines and Poisons**

No poison schedule number allocated.

| Component                        | Standard for the Uniform Scheduling of Medicines and Poisons |
|----------------------------------|--------------------------------------------------------------|
| Butyl benzyl phthalate - 85-68-7 | Schedule 10 listed                                           |

**Australian Industrial Chemicals Introduction Scheme (AICIS)**

| Component                        | Australian Industrial Chemicals Introduction Scheme (AICIS) | Additional information                                                                                                                                                                                                       |
|----------------------------------|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Butyl benzyl phthalate - 85-68-7 | Present                                                     | Specific information requirement: Obligations to provide information apply. You must tell us within 28 days if the circumstances of your importation or manufacture (introduction) are different to those in our assessment. |

**Australian - Illicit Drug Precursors/Reagents Substance List**

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.

**Chemicals of Security Concern**

This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

**National pollutant inventory**

Not applicable

**Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

**International Inventories**

| Component              | AICS | NZIoC | EINECS    | ELINCS | TSCA | DSL | NDSL | PICCS | ENCS | ISHL | IECSC | KECL     |
|------------------------|------|-------|-----------|--------|------|-----|------|-------|------|------|-------|----------|
| Butyl benzyl phthalate | X    | X     | 201-622-7 | -      | X    | X   | -    | X     | X    | X    | X     | KE-02200 |

**Legend:** X - Listed. '-' - Not Listed. **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**International Regulations****Ozone Depletion Potential**

This product does not contain any known or suspected substance

**Persistent Organic Pollutant**

This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)**

Not applicable

**MARPOL** - International Convention for the



## Prevention of Pollution from Ships

| Component                        | IMDG Marine Pollutant                                 |
|----------------------------------|-------------------------------------------------------|
| Butyl benzyl phthalate - 85-68-7 | IMDG regulated marine pollutant (Listed in the index) |

## Basel convention on the control of transboundary movements of hazardous wastes and their disposal

Not applicable.

| Component              | CAS No  | OECD HPV | Restriction of Hazardous Substances (RoHS) | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|------------------------|---------|----------|--------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Butyl benzyl phthalate | 85-68-7 | Listed   | 0.1% (Max. Conc.)                          | Not applicable                                                                            | Not applicable                                                                           |

## Authorisation/Restrictions according to EU REACH

| Component              | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization                                                                                                                                                                                                                     | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                                                                                            | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)                              |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Butyl benzyl phthalate | Toxic for reproduction Category 1B<br>Article 57<br>Application date: August 21, 2013<br>Sunset date: February 21, 2015<br>Exemption - Uses in the immediate packaging of medicinal products covered under Regulation (EC) No 726/2004 Directive 2001/82/EC and/or Directive 2011/83/EC | Use restricted. See item 30. (see link for restriction details)<br>Use restricted. See item 51[c]. (see link for restriction details)<br>Use restricted. See item 75. (see link for restriction details) | SVHC Candidate list - 201-622-7 - Toxic for reproduction, Article 57c; Endocrine disrupting properties, Article 57f - human health |

After the sunset date the use of this substance requires either an authorization or can only be used for exempted uses, e.g. use in scientific research and development which includes routine analytics or use as intermediate.

<https://echa.europa.eu/authorisation-list>

<https://echa.europa.eu/substances-restricted-under-reach>

<https://echa.europa.eu/candidate-list-table>

## Section 16 - Other Information

## Legend

**AICS** - Australian Inventory of Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**IECSC** - Chinese Inventory of Existing Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**NZS 5433:2012** - Transport of Dangerous Goods on Land

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**WEL** - Workplace Exposure Limit

**DNEL** - Derived No Effect Level

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**VOC** - (Volatile Organic Compound)

**NZIoC** - New Zealand Inventory of Chemicals

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**CAS** - Chemical Abstracts Service

**ACGIH** - American Conference of Governmental Industrial Hygienists  
Predicted No Effect Concentration (PNEC)

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**ADG** Australian Code for the Transport of Dangerous Goods by Road and Rail

**OECD** - Organisation for Economic Co-operation and Development

**LC50** - Lethal Concentration 50%

**ATE** - Acute Toxicity Estimate

**RPE** - Respiratory Protective Equipment

**NOEC** - No Observed Effect Concentration

**BCF** - Bioconcentration factor

**PBT** - Persistent, Bioaccumulative, Toxic

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

**Revision Date**

18-Nov-2022

**Revision Summary**

Not applicable.

**This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).**

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**